

An Empirical Study on the Relationship between the Use of Agile Practices and the Success of Scrum Projects

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ABSTRACT

In this article, factors considered critical for the success of projects managed using Scrum are correlated to the results of software projects in industry. Using a set of 25 factors compiled in by other researchers, a cross section survey was conducted to evaluate the presence or application of these factors in 11 software projects that used Scrum in 9 different software companies located in Recife-PE, Brazil. The questionnaire was applied to 65 developers and Scrum Masters, representing 75% (65/86) of the professionals that have participated in the projects. The result was correlated with the level of success achieved by the projects, measured by the subjective perception of the project participant, using Spearman's rank correlation coefficient. The main finding is that only 32% (8/25) of the factors correlated positively with project success, raising the question of whether the factors hypothesized in the literature as being critical to the success of agile software projects indeed have an effect on project success. Given the limitations regarding the generalization of this result, other forms of empirical results, in particular case-studies, are needed to test this question.

Categories and Subject Descriptors

D.2.9 [Software Engineering]: Management – *programming teams*.

General Terms

Management, Performance, Experimentation, Human Factors.

Keywords

Scrum, Project success, Agile practices, Survey, Empirical study.

1. INTRODUCTION

Scrum [17][18] has been gaining attention among software development organizations, specially by those seeking to increase the success rate of their projects through adopting agile methodologies, process and practices.

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According to Highsmith [10], the word “success” can assume many different meanings depending on individuals and context in which it is discussed. Within the context of agile methods, project success has been treated as the effective adoption of a pre-conceived set of agile practices [1][2][4][5][6][10][13][14][15]. However, it has already passed nine years from the Agile Manifesto [2] publication and there are still only little empirical evidence on the actual effect of adopting agile practices on the project success as a whole [5].

Chow and Cao [5] conducted a survey of 109 agile projects from 25 different countries. Using multiple regression techniques, they found six critical success factors hypothetically related to agile project success: Team Environment, Team Capability, Customer Involvement, Project Management Process, Agile Software Engineering Techniques, and Delivery Strategy. They also described these critical success factors in terms of 25 agile attributes. These results are important because they show that certain agile practices are related to project success, offering guidance for practitioners that seek to improve success rates in software projects.

In this work, our goal is to test the hypothesis raised by Chow and Cao [5] in the context of software companies that use Scrum for project management and are located in the City of Recife, Brazil. We chose to investigate Scrum due to its increasing adoption in industry, in particular among small and medium companies in our context. Therefore, we want to answer the following research questions:

- *RQ1: Is the adoption of the 25 agile attributes described by Chow and Cao [5] related to the success of software development projects managed using Scrum?*

In the rest of this article, we describe a survey performed with 62 software engineers, covering engineers who have worked in Scrum projects and also Scrum Masters, associated to 11 projects in 9 different software firms from Recife-PE, Brazil. The participants answered a closed questionnaire on their own perception about the general success of the Scrum project and also on their own perception about the adoption level of each of the 25 agile attributes described by Chow and Cao [5].

The correlations of these agile attributes to project success were tested, and the results show that only eight out of the 25 attributes are positively correlated to the success of the evaluated software projects. Then, we performed a Principal Component Analysis in order to reduce the 25 agile attributes into a smaller and non-

interrelated group of factors. Finally, we compare our results to those discussed in Chow and Cao [5].

These results cast doubt on the efficacy of adopting certain practices advocated in the literature to increase the probability of project success. However, the limited context and small number of projects represent a threat to external validity. Besides, the use of survey to test for causal relations is also limited. Other forms of empirical studies are necessary to further investigate this research problem. Case studies [15] in industrial settings seem to be a reasonable research method for future research.

The rest of this article is structure as follows. In Section 2, some related work is briefly described. In Section 3, the research method is described. In Section 4, the results are presented. Finally, in Section 5, conclusions are presented, and some directions for future research are proposed.

2. RELATED WORK

Scrum is described as an agile framework, composed of a set of agile practices and roles, used for managing and controlling systems and products development, following an iterative and incremental paradigm [17][20]. Schwaber [17] summarizes the basic principles of Scrum as: small working teams, willingness to the change, and short delivery of shippable products. These principles are indeed aligned to those of the Agile Manifesto [2].

In 2007, Chow and Cao [5] performed a research aiming at evaluating the relationship of software development projects success goals (defined in terms of quality, scope, schedule, and cost) with a set of agile practices described on the Agile Manifesto [2]. From a survey covering 109 companies in 25 countries, and based on statistical analysis, they came up with a list of 12 factors that were posed as hypothetical critical success factors to generic software development agile projects. After deeper analysis, only 6 factors were considered as actually related to agile software development projects success: delivery strategy, agile software engineering techniques, team capability, project management process, team environment and customer involvement. In order to enable the verification of their claim about the six critical success factors, Chow and Cao [5] also described a set of observable project attributes corresponding to the operationalization of those six factors (see in Table 1).

Misra [14] tried to identify critical success factors in adopting agile software development practices by means of an *ex-post-factum* research. Several other researchers have postulated that success in adoption of the agile practices is equivalent to project success [1][2][4][5][6][10][13][14][15]. However, this postulate still requires empirical confirmation.

3. RESEARCH METHOD

This research uses a quantitative approach supported by statistical methods. The observed phenomenon is the use adoption of the 25 agile attributes described by Chow and Cao [5] in software projects managed using Scrum. Our purpose is to evaluate the relationship between the adoption of the attributes and project success, the perspective of the software team.

3.1 Survey Design

This research was carried out in four steps:

Step I – Background Knowledge: Relevant research on agile practices, in particular the results of Chow and Cao [5], were studied in depth in order to consolidate the necessary conceptual background for the next steps.

Table 1 - Critical Success Factors and Agile Attributes
(source: Chow and Cao [5])

Critical Success Factors	Agile Attributes
Delivery strategy	A ₀₁ Regular delivery of software
	A ₀₂ Delivering most important features first
Agile software engineering techniques	A ₀₃ Well-defined coding Standards up front
	A ₀₄ Pursuing simple design
	A ₀₅ Rigorous refactoring activities
	A ₀₆ Right amount of documentation
	A ₀₇ Correct integration testing
Team capability	A ₀₈ Team members with high competence and expertise
	A ₀₉ Team members with great motivation
	A ₁₀ Managers knowledgeable in agile
	A ₁₁ Managers who have adaptative management style
	A ₁₂ Appropriate technical training to team
Project management process	A ₁₃ Following agile-oriented requirement management process
	A ₁₄ Following agile-oriented project management process
	A ₁₅ Following agile-oriented configuration management process
	A ₁₆ Good progress tracking mechanism
	A ₁₇ Strong communication focus with daily face-to-face meetings
	A ₁₈ Honoring regular working Schedule
Team environment	A ₁₉ Collocation of the whole team
	A ₂₀ Coherent, self-organizing team-work
	A ₂₁ Projects with small team
Customer involvement	A ₂₂ Projects with no multiple independent teams
	A ₂₃ Good customer relationship
	A ₂₄ Strong customer commitment and presence
	A ₂₅ Customer having full authority

Step II – Study Design: A survey questionnaire was designed to evaluate the level of adoption of the 25 agile attributes of Table 1 and the project success from the software engineers' perspective. Finally, a survey was planned involving software engineers from software companies located in Recife, PE (Brazil). The participant companies were selected initially from the list of organizations which participate in the Recife's Scrum user group. Parctitioners from these organizations were then asked for other companies using Scrum. From the final list of 20 organizations, 10 were discarded because they had not finished project managed using Scrum or were not using Scrum in software development.

Step III – Study Development: the data collection occurred in 2009, between 27th of July and 07th of August. A total of 86 questionnaires were distributed. Based on previous experience [8], we decided to distribute printed questionnaires in order to achieve a high response rate.

Step IV - Data analysis and synthesis: First, the agile attributes were correlated to project success using Spearman's rank correlation coefficient. Then, a Principal Component Analysis (PCA) [9] was run in order to identify a grouping of factors that could explain the underlying structure of the agile attributes. Finally, we compared our results to Chow and Cao [5].

3.2 Operational Definition of Variables

The variable *The Adoption of Agile Attributes* was operationalized by means of the perception of software engineers about the level of adoption of each attribute (A₀₁... A₂₅, from Table 1). In order to measure the level of adoption a 5 point Likert type scale was

used with the following options: totally disagree (TD), partially disagree (PD), neutral (N), partially agree (PA), totally agree (TA).

The variable *Project Success* was operationalized by means of the perception of the software engineers about the general success of the project (S), also using a 5 point Likert scale with the following options: failure (F), almost failure (AF), partial success (PS), almost total success (ATS), total success (TS). Our questionnaire was pre-tested with four Scrum masters, who answered and reviewed the questions. The answers to the pre-test did enter in the final results. The scales were later converted to an ordinal scale (0...4) for the use of the statistical correlation method.

3.3 Sampling Method and Data Analysis

Participants were randomly sampled from these 10 companies selected in Step II. All participants signed an Informed Consent Form. Late, blank and obliterated questionnaires were discarded from analysis. The data analysis was carried out using SPSS® Statistics Package software, version 17¹.

4. RESULTS AND DISCUSSIONS

A total of 62 valid questionnaires were returned from the 65 distributed one, achieving 95% of response rate, which rules out self selection problems. This corresponds to 72% (62/86) of the total population from the studied projects. The majority of the interviewed engineers, 51 (82%), are male, 54 (84%) are aged from 20 to 30 years old, 29 (47%) has a high education degree and 21 (34%) have post-graduate degree. The research covered the three Scrum team roles: 10 Scrum masters (16%), 4 product owners (6%), 48 (78%) team members (analysts, designers, developers and testers).

The general frequency distribution of *Project Success* and *The Adoption of Agile Attributes* are shown in Table 2 and Table 3, respectively.

In Table 4, the results of the Spearman's rank correlation coefficient between the two variables are presented. According to Table 4, only eight of the 25 attributes presented a significant correlation to project success: A₀₁ - *Regular delivery of software*, A₀₂ - *Delivering most important features first*, A₀₇ - *Correct integration testing*, A₀₈ - *Team members with high competence and expertise*, A₁₃ - *Following agile-oriented requirement management process*, A₁₅ - *Following agile-oriented configuration management process*, A₂₀ - *Coherent, self-organizing team-work*, A₂₃ - *Good customer relationship*. This result agrees with some of Scrum basic principles, such as short Sprints and prioritizing most valuable features. Additionally, this result is consistent with a set of current studies on the influence of human factors on software engineering projects [8]. However, not all attributes describe Chow and Cao [5], as would be expected.

Table 2 - Project Success

F	AF	PS	ATS	TS
0 (0%)	0 (0%)	4 (6%)	26 (42%)	32 (52%)

Table 3 – Use of agile practices

TD	PD	N	PA	TA
69 (4.5%)	137 (8.8%)	217 (14%)	472 (30.5%)	655 (42.3%)

Table 4 - Agile attributes and project success correlation test

	Project Success (S)	
	Correlation Coefficient	Sig.
A ₀₁	.441**	.000
A ₀₂	.306*	.016
A ₀₃	.165	.200
A ₀₄	.117	.366
A ₀₅	.120	.354
A ₀₆	.211	.099
A ₀₇	.316*	.012
A ₀₈	.283*	.026
A ₀₉	.154	.232
A ₁₀	.228	.074
A ₁₁	.136	.291
A ₁₂	.135	.295
A ₁₃	.298*	.019
A ₁₄	.184	.153
A ₁₅	.326*	.010

	Project Success (S)	
	Correlation Coefficient	Sig.
A ₁₆	.034	.791
A ₁₇	.238	.063
A ₁₈	.246	.054
A ₁₉	-.150	.244
A ₂₀	.322*	.011
A ₂₁	-.058	.656
A ₂₂	.038	.768
A ₂₃	.316*	.012
A ₂₄	.083	.521
A ₂₅	-.191	.137

*. Correlation is significant at the 0.05 level (2-tailed).
 ** Correlation is significant at the 0.01 level (2-tailed).

We then performed a principal component analysis (PCA) on the attributes. The result of the KMO test was 0.74, which means that the data factored well [9]. Table 5 (5.1 and 5.2) shows the attributes grouped according to the nine components found as the result of the PCA, based on their higher loadings. Names have been suggested to the principal components trying to summarize the meaning of all or most of the attributes with significant loads in the component (*load* > 0.4).

The second finding in this study is that this component grouping did not match exactly with the Critical Success Factor described by Chow and Cao [5] (Table1), and the main difference are summarized here. The factors C5 - *Team capability*, and C7 - *Delivery strategy*, received the same names in both studies, even though they do not comprehend exactly the same attributes. Team environment [5] seems to have been divided in two factors here: C5 - *Team structure*, and C8 - *Team location*. Project management process [5] seems to have been divided in two factors here: C1 - *Management Style* and C2 - *Software process*. And Customer Involvement [5] seems to have been divided in two other factors here: C6 - *Customer commitment*, and C9 - *Customer awareness*. Agile Software Engineering Techniques [5] also has some attributes in common with our C4 - *Technology*.

Table 5.1(a) - Components definition

C1 Management style	A ₁₇ Strong communication focus with daily face-to-face meetings
	A ₁₀ Managers knowledgeable in agile
	A ₁₈ Honoring regular working Schedule
	A ₁₁ Managers who have adaptative management style
	A ₂₀ Coherent, self-organizing team-work
C2 Software process	A ₂₃ Good customer relationship
	A ₁₃ Following agile-oriented requirement management process
	A ₁₅ Following agile-oriented configuration management process
	A ₁₄ Following agile-oriented project management process
C3 Team Structure	A ₀₂ Delivering most important features first
	A ₂₂ Projects with no multiple independent teams
	A ₂₁ Projects with small team

¹ <http://www.spss.com/software/statistics>

Table 5.1(b) - Components definition

C4 Technology	A ₀₃ Well-defined coding Standards up front A ₀₇ Correct integration testing A ₁₂ Appropriate technical training to team A ₀₄ Pursuing simple design
C5 Team capability	A ₀₉ Team members with great motivation A ₀₈ Team members with high competence and expertise
C6 Customer commitment	A ₂₄ Strong customer commitment and presence A ₀₆ Right amount of documentation
C7 Delivery strategy	A ₀₁ Regular delivery of software
C8 Team Location	A ₁₉ Collocation of the whole team A ₀₅ Rigorous refactoring activities
C9 Customer awareness	A ₂₅ Customer having full authority A ₁₆ Good progress tracking mechanism

5. CONCLUSIONS AND FUTURE WORK

This work aimed at investigating the relationship between the use of agile practices and the success of software projects managed using Scrum. A cross-sectional survey was performed, covering 62 software engineers who had worked among 11 software development Scrum projects from 9 different organizations in Recife-PE, Brazil.

Two results have been found. First, only 8 out of 25 agile attributes described by Chow and Cao [5] correlate to project success. Two important implications of this result should be further investigated. On the one hand, it is possible that some attributes normally assumed to have effect on project success actually do not have. On the other hand, other “non-agile” attributes of project management can be influencing project success, and have not been treated by Chow and Cao [5] nor in our research.

In the second result, we grouped the 25 agile attributes in a set of nine higher level factors using principal component analysis. These factors group attributes with similar intuitive meaning, as can be seen by the labels used to name the nine factors. Although the components do not exactly match the Critical Success Factor described by Chow and Cao [5], the grouping is not inconsistent with those factors.

This work has some limitations and threats to validity. First, the small and localized set of participant firms and engineers represent a threat to external validity, i.e., the results cannot be generalized outside the context of our study. Second, we have not tested the psychometric properties of the survey questionnaire. Since it is a personal perception type of questionnaire, not testing its psychometric properties threatens construct validity. Third, our operationalization of the variables, mainly Project Success, is different from Chow and Cao [5]. Therefore, although we can compare both results at the theoretical level, direct comparisons cannot be performed.

Considering the results and the limitations of this research, we believe that alternative research methods should be used to provide a deeper understanding of the influence of agile practices on project success. In particular, case studies [16] could provide “thick descriptions” (a term popularized by Geertz [7]) of the phenomenon of interest. Our research team is currently designing a series of case studies with this objective, based on the questions that have been raised by the work reported here. We hope that

other researchers would be interested in replicating these studies with the objective of contributing towards theoretical foundations to understand the effects of the agile methods in practice.

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